

- 3.31** Data on weekly sales of a major brand of canned tuna by a supermarket chain in a large midwestern U.S. city during a mid-1990s calendar year are contained in the data file *tuna*. There are 52 observations for each of the variables. The variable $SAL1$ = unit sales of brand no. 1 canned tuna, and $APR1$ = price per can of brand no. 1 tuna (in dollars).
- Calculate the summary statistics for $SAL1$ and $APR1$. What are the sample means, minimum and maximum values, and their standard deviations. Plot each of these variables versus $WEEK$. How much variation in sales and price is there from week to week?
 - Plot the variable $SAL1$ (y-axis) against $APR1$ (x-axis). Is there a positive or inverse relationship? Is that what you expected, or not? Why?
 - Create the variable $PRICE1 = 100APR1$. Estimate the linear regression $SAL1 = \beta_1 + \beta_2 PRICE1 + e$. What is the point estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1? What is a 95% interval estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1?
 - Construct a 90% interval estimate for the expected number of cans sold in a week when the price per can is 70 cents.
 - Construct a 95% interval estimate of the elasticity of sales of brand no. 1 with respect to the price of brand no. 1 “at the means.” Treat the sample means as constants and not random variables. Do you find the sales are fairly elastic, or fairly inelastic, with respect to price? Does this make economic sense? Why?
 - Test the hypothesis that elasticity of sales of brand no. 1 with respect to the price of brand no. 1 from part (e) is minus three against the alternative that the elasticity is not minus three. Use the 10% level of significance. Clearly, state the null and alternative hypotheses in terms of the model parameters, give the rejection region, and the p -value for the test. What is your conclusion?

a.

SAL1:

sample means : 6719

minimum value : 1762

maximum values : 32820

standard deviation : 6915.083

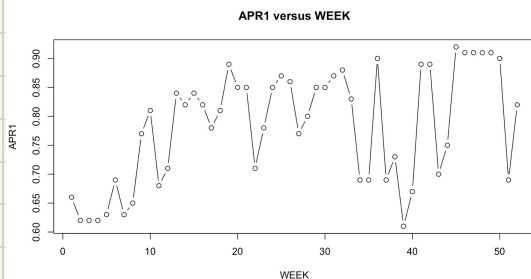
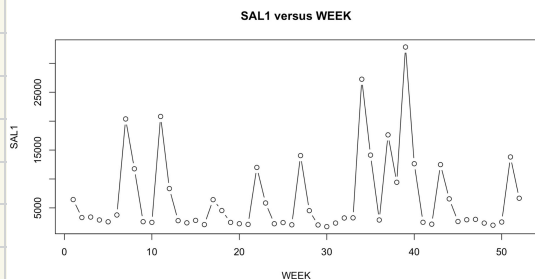
APR1:

sample means : 0.7825

minimum value : 0.6100

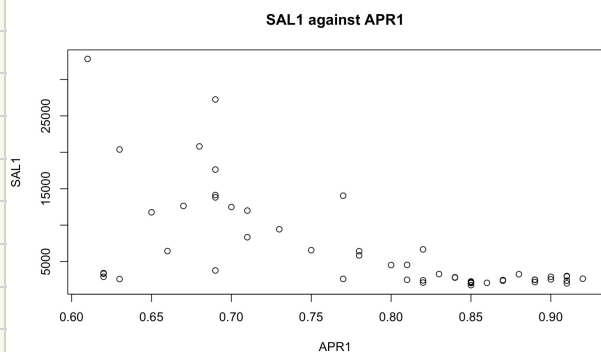
maximum values : 0.9200

standard deviation : 0.09803711



There is a large amount of week-to-week variation in sales, but only moderate variation in price.

b.



The scatter plot suggests an inverse relationship between SAL1 and APR1.

As the price of brand no.1 tuna increases, sales tend to decrease.

This is what we would expect from the law of demand: higher prices generally reduce quantity demand, while lower prices increase sales.

c.

$$SAL1 = \beta_1 + \beta_2 PRICE1 + e$$

$$\frac{dSAL1}{dPRICE1} = \beta_2$$

$$\hat{\beta}_2 = -434.4473$$

$$95\% \text{ C.I. for } \beta_2 : [-592.2897, -276.6049]$$

d. 90% C.I. for $E[SAL1|PRICE1=70]: [8624.934, 11980.87]$

e.

$$\begin{aligned}\text{elasticity} &= \frac{\Delta SAL1}{\Delta APR1} \cdot \frac{APR1}{SAL1} \\&= \frac{1}{100} \cdot \frac{\Delta SAL1}{\Delta PRICE1} \cdot \frac{PRICE1}{SAL1} \\&= 1.2 \cdot \frac{PRICE1}{SAL1}\end{aligned}$$

95% C.I. of the elasticity: $[-6.898149, -3.221501]$

The sales of brand no.1 are price elastic at the means, since the absolute value of elasticity is greater than 1 throughout the interval.

This is economically plausible because consumers may respond strongly to tuna prices when close substitutes are available.

f.

① $H_0: E = -3$

$H_A: E \neq -3$

② $\alpha = 0.10$

③ test statistic: $\frac{\hat{E} - (-3)}{se(\hat{E})} H_0 \sim t_{(52-2)}$

④ $RR = \{ |t| \geq t_{0.05, 50} = 1.6759 \}$

⑤ $t_0 = -2.250572$

⑥ $p\text{-value} = 2 \cdot P(t \leq -2.250572 | H_0 \text{ True}) = 0.02884204$

⑦ $p\text{-value} \leq 0.1$, reject H_0

We have sufficient evidence to indicate that elasticity of sales of brand no.1 with respect to the price of brand no.1 at the means is significantly different from -3 at the 10% significance level.